

PRE-DRIVE PRODUCT ENGINEERING ASSIGNMENTS

Walk-in Recruitment Drive • Candidate Assessment Brief • Choose ONE of the five assignments

Important: Choose **ONE** of the five business problems. There is no single expected solution. We evaluate problem understanding, product thinking, assumptions, engineering decisions, solution quality, validation, and ability to explain your work.

General Instructions

- Choose one assignment from the five options.
- The problem statements intentionally do not prescribe detailed functional requirements, UI designs, technology choices, database structures, APIs, workflows, or architecture. Decide these yourself.
- **AI-assisted development is allowed and encouraged.** You may use ChatGPT, Claude, Gemini, Cursor, Copilot, or other AI-assisted development tools.
- Complete the **mandatory AI Usage Report** and clearly explain how AI was used and how you validated its output.
- Submit a working solution/prototype, source code/repository, brief approach note, assumptions, and AI/tool usage details.

Assignment 1 — Smart Attendance Management

Business Problem

A college has approximately 5,000 students, 200 faculty members, multiple departments, classes, sections, and subjects. Build a solution that helps the institution manage attendance efficiently.

Your Challenge

Address attendance recording, corrections, review, history, and identification of students with low attendance. Decide users, workflows, permissions, reports, UX, data model, architecture, and implementation.

Assignment 2 — Fee Collection & Reconciliation

Business Problem

Design a solution for managing student fee collection across multiple fee heads, discounts/concessions, installments, payments, failures, reversals, outstanding amounts, and payment history.

Your Challenge

Think beyond recording payments. Consider financial consistency, reconciliation, auditability, edge cases, user roles, reporting, and uncertain or changing payment states.

Assignment 3 — Intelligent Timetable Generator

Business Problem

A college needs to generate timetables across multiple divisions, subjects, faculty members, classrooms, and periods.

Your Challenge

Build a solution that generates and manages timetables while respecting hard constraints such as teacher/classroom/division conflicts, room requirements, and scheduling relationships. Consider impossible or conflicting constraints and how users are informed.

Assignment 4 — Student Support & Ticket Management

Business Problem

Students raise requests related to fees, attendance, ID cards, documents, certificates, and other administrative matters. Staff need to own, prioritize, process, and resolve these requests.

Your Challenge

Design a ticket/support system with statuses, priorities, assignment, SLAs, ageing, ownership, resolution tracking, activity history, and management visibility. Decide escalation and pending-action workflows.

Assignment 5 — Admission Lead Management

Business Problem

An educational institution receives admission leads through its website, walk-ins, phone calls, WhatsApp, fairs, campaigns, and other sources.

Your Challenge

Build a solution to manage the lead lifecycle from first contact through follow-up and conversion. Consider counsellor assignment, course preferences, lead sources, status, follow-up actions, manager visibility, ageing, and useful reports/insights.

Submission Expectations

- Working solution/prototype
- Source code/repository
- Brief explanation of approach, assumptions, architecture, and trade-offs
- Validation and important edge cases
- Mandatory AI/tool usage report

Mandatory AI Usage Report

AI TOOL USED: ChatGPT / Claude / Gemini / Cursor / Copilot / Other

WHAT I ASKED AI TO DO:

1. _____
2. _____
3. _____

PROMPT THAT WAS MOST USEFUL:

CODE GENERATED BY AI: What part? _____

CODE I MODIFIED: What part? _____

AI OUTPUT THAT WAS WRONG:

HOW I IDENTIFIED THE PROBLEM:

HOW I FIXED IT:

Submission Details

Deadline	9:00 AM, 25 September 2026
Submission Email	tech_interview@edumerge.com

Important — Walk-in Drive Shortlisting

Submission of the assignment does **not** confirm participation in the walk-in drive. All submissions will be reviewed by the technical/product team. Shortlisted candidates will receive a **separate confirmation email** with walk-in details.

Only candidates who receive the shortlisting/confirmation email should attend the walk-in drive. Candidates who do not receive a confirmation email should not report to the venue.

Evaluation Areas

- Problem understanding
- Product thinking and assumptions
- Technical approach and engineering decisions
- Solution quality and usability
- Edge cases and failure scenarios
- Creativity and practicality
- AI/tool usage and validation
- Ability to explain implementation decisions